

SARTHAK TANDEL

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EDUCATION

Dharmsinh Desai University

Bachelor of Technology in Information Technology — CPI: 8.58

Nadiad, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, C/C++, SQL

Core Technologies: Spring Boot, Hibernate, JPA, React.js, Node.js, WebRTC, WebSockets, gRPC, Kafka

Databases: MySQL, PostgreSQL, Redis (Caching)

Cloud and DevOps: AWS, Docker, Kubernetes, GitHub Actions

Concepts: System Design, Microservices, Distributed Systems, OOP, REST APIs, DBMS

PROJECTS

Studify | *React, Spring Boot, PostgreSQL, WebRTC, WebSockets, Docker, AI Integration*

- Created a scalable real-time online learning platform supporting live video/audio classes and screen sharing.
- Built **real-time meeting and broadcast streaming features** with low-latency communication using **WebRTC SFU architecture** and event-driven participant synchronization.
- Containerized the entire application stack with **Docker** to enable reproducible deployments, and **deployed it on AWS** to achieve scalability, high availability, and efficient resource management.
- Developed AI-powered classroom tools including PDF summarization and an intelligent classroom assistant, reducing manual content-review time for instructors.

IrisPay | *React, Spring Boot, Flask, DeepFace, ArcFace, RetinaFace, Microservices*

- Architected a microservices-based biometric payment authentication system with a React frontend, Spring Boot backend, and a Flask ML inference service for face recognition
- Integrated **ArcFace and RetinaFace** via DeepFace to enable high-accuracy face detection and embedding generation for real-time identity verification
- Implemented face **vectorization with cosine similarity** for authentication, incorporating **multi-face detection and strict thresholding for fraud prevention**.

Qnect | *React, Spring Boot, PostgreSQL, JWT, WebSockets, AI Integration*

- Built a full-stack Q&A platform integrating an AI-powered answer engine to deliver instant, context-aware responses, reducing user query resolution time.
- Implemented bidirectional **real-time communication via WebSockets**, enabling live chat, targeted answer requests, and push-based notifications across concurrent users.
- Designed and secured RESTful APIs using Spring Boot with stateless JWT authentication; modeled relational data in PostgreSQL using Hibernate/JPA.
- Integrated **Redis caching** and real-time news APIs to optimize response latency, reduce database load, and surface trending topics for increased user engagement.

CodeIntel | *Spring Boot, FastAPI, Redis, Kafka, AST Parsing, Docker*

- Built a **microservices-based platform** using **Spring Boot** and **FastAPI** for API routing, query handling, and code analysis.
- Developed an **AST-based code ingestion pipeline** using **Tree-sitter** and **Apache Kafka** to process and index **multi-language codebases** (Java, Python, JavaScript/TypeScript).
- Implemented a Vectorless **RAG pipeline using BM25 lexical search, AST-derived dependency graphs**, and Redis caching, enabling fast and context-aware code retrieval.
- Secured and deployed the platform with **JWT authentication, Bucket4j rate limiting, and Dockerized services** including databases and **Nginx** servers.

ACHIEVEMENTS & CERTIFICATIONS

DUHacks 5.0 Hackathon — Ranked in the **Top 10** at DUHacks 5.0 Hackathon for building **IrisPay**, an AI-powered biometric payment authentication system.

AWS Academy Cloud Foundations — Completed AWS Academy course covering core cloud computing concepts, AWS services, security, architecture, pricing, and support.

LeetCode — Solved 300+ problems across data structures and algorithms topics.